

Detailed Scene Setup Guide

This guide walks you through manually setting up the Unity scenes.

Setting Up the Main Menu Scene

Step 1: Create New Scene

1. Open Unity
2. Go to **File** → **New Scene** → **Empty Scene**
3. Save immediately as `Assets/Scenes/MainMenu.unity`

Step 2: Create Camera

1. Right-click in Hierarchy → **Camera**
2. Name it "Main Camera"
3. Tag it as "MainCamera"
4. Set these properties:
 - Clear Flags: Solid Color
 - Background: Dark blue (R:13, G:13, B:38)
 - Projection: Orthographic
 - Size: 5

Step 3: Create Scene Manager

1. Right-click in Hierarchy → **Create Empty**
2. Name it "SceneManager"
3. Click **Add Component** → search "MainMenuSetup"
4. Enable "Setup On Start" checkbox

Step 4: Create GameManager (Persistent)

1. Right-click in Hierarchy → **Create Empty**
2. Name it "GameManager"
3. Click **Add Component** → search "GameManager"

Step 5: Save Scene

- Press **Ctrl+S** to save

Setting Up the Game Scene

Step 1: Create New Scene

1. Go to **File** → **New Scene** → **Empty Scene**
2. Save immediately as `Assets/Scenes/Game.unity`

Step 2: Create Camera

1. Right-click in Hierarchy → **Camera**
2. Name it "Main Camera"

3. Tag it as “MainCamera”
4. Set properties:
 - Clear Flags: Solid Color
 - Background: Dark blue (R:13, G:13, B:38)
 - Projection: Orthographic
 - Size: 5

Step 3: Create Scene Setup

1. Right-click in Hierarchy → **Create Empty**
2. Name it “SceneSetup”
3. Add Component → “SceneSetup”
4. Check all boxes:
 - ☒ Setup On Start
 - ☒ Create Player
 - ☒ Create UI
 - ☒ Create Background
 - ☒ Create Spawners

Step 4: Create Prefab Generator

1. Right-click in Hierarchy → **Create Empty**
2. Name it “PrefabGenerator”
3. Add Component → “PrefabGenerator”

Step 5: Save Scene

- Press **Ctrl+S** to save

Alternative: Manual Object Setup

If you prefer to set up objects manually instead of using the auto-setup scripts:

Create Player Manually

1. **Create Player Object**
 - Right-click Hierarchy → **2D Object** → **Sprite** → **Square**
 - Name it “Player”
 - Tag it as “Player”
 - Position: (0, -3.5, 0)
2. **Add Components**
 - Add Component → **Box Collider 2D**
 - Is Trigger: ☒
 - Size: (0.8, 0.8)
 - Add Component → **Rigidbody 2D**
 - Gravity Scale: 0
 - Constraints → Freeze Rotation: ☒
 - Add Component → **PlayerController**

3. Create Fire Point

- Right-click on Player → **Create Empty**
- Name it "FirePoint"
- Position: (0, 0.5, 0)

4. Create Shield Visual

- Right-click on Player → **2D Object** → **Sprite** → **Circle**
- Name it "ShieldVisual"
- Scale: (2, 2, 1)
- Color: Cyan with low alpha
- Disable it (uncheck the checkbox)

Create Bullet Prefab

1. Create Bullet Object

- Right-click Hierarchy → **2D Object** → **Sprite** → **Square**
- Name it "PlayerBullet"
- Tag it as "PlayerBullet"
- Scale: (0.15, 0.3, 1)
- Color: Cyan

2. Add Components

- Add Component → **Box Collider 2D**
 - Is Trigger: ☒
 - Add Component → **Rigidbody 2D**
 - Gravity Scale: 0
 - Add Component → **Bullet**

3. Create Prefab

- Drag from Hierarchy into `Assets/Prefabs/Bullets/`
- Delete from scene

Create Enemy Prefab

1. Create Enemy Object

- Right-click Hierarchy → **2D Object** → **Sprite** → **Square**
- Name it "BasicEnemy"
- Tag it as "Enemy"
- Color: Red

2. Add Components

- Add Component → **Box Collider 2D** (Is Trigger: ☒)
- Add Component → **Rigidbody 2D** (Gravity Scale: 0)
- Add Component → **BasicEnemy**

3. Create Prefab

- Drag into `Assets/Prefabs/Enemies/`
- Delete from scene

Wire Up References

After creating prefabs, connect them:

1. Select **PlayerController** on Player

2. Drag **PlayerBullet** prefab to “Bullet Prefab” field
 3. Drag **FirePoint** to “Fire Points” array
 4. Select **EnemySpawner**
 5. Drag enemy prefabs to their respective fields
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Build Settings Configuration

1. Go to **File → Build Settings**
 2. Click **Add Open Scenes** or drag scenes:
 - MainMenu (should be index 0)
 - Game (should be index 1)
 3. Platform Settings:
 - Platform: **PC, Mac & Linux Standalone**
 - Target Platform: **Windows**
 - Architecture: **x86_64**
 4. Click **Switch Platform** if needed
 5. Click **Player Settings** and configure:
 - Product Name: Space Shooter
 - Resolution: 1920 x 1080
 - Fullscreen Mode: Windowed (or Fullscreen Window)
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Testing the Game

1. Open the MainMenu scene
2. Press **Play** button
3. Click “Start Game” to begin
4. Test controls:
 - WASD/Arrows to move
 - Space to shoot
 - ESC to pause

Common Setup Issues

“Script not found” error:

- Make sure scripts are in Assets/Scripts folder
- Check script names match class names
- Wait for Unity to finish compiling

Player doesn’t move:

- Verify the PlayerController component is attached
- Check Edit → Project Settings → Input Manager
- Make sure Rigidbody2D has Freeze Rotation enabled

Bullets don't appear:

- Verify bullet prefab is assigned to PlayerController
- Check that bullet prefab has Bullet script attached
- Ensure FirePoint exists as child of Player

Enemies don't spawn:

- Check EnemySpawner has enemy prefabs assigned
- Verify GameManager state is "Playing"
- Make sure enemy prefabs have correct Enemy scripts